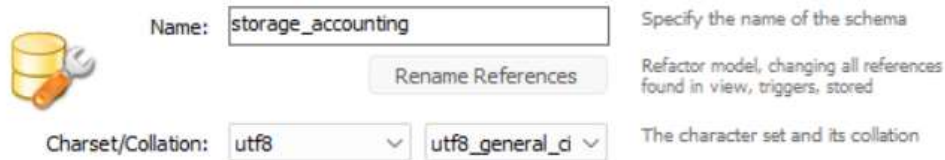


Лабораторная работа №6

Инициализация среды для работы с базами данных MySQL, запуск docker контейнера storage_accounting с образом MySQL.

```
C:\Users\diamo>docker run --name storage_accounting -v C:\mysql:\var\lib\mysql -p 3306:3306 -e MYSQL_ROOT_PASSWORD=my-secret-pw -d mysql:8.0.36 da74638383171caaf9487bec41cc6a47ec31e8caa6e0f805445940644701327d
```



The image shows a screenshot of the MySQL Workbench Schema Designer. On the left, there is a yellow cylinder icon representing a database. The main area contains the following fields and buttons:

- Name:** A text box containing 'storage_accounting'. To its right is the text 'Specify the name of the schema'.
- Rename References:** A button with the text 'Rename References'. To its right is the text 'Refactor model, changing all references found in view, triggers, stored'.
- Charset/Collation:** Two dropdown menus. The first is set to 'utf8' and the second is set to 'utf8_general_ci'. To the right of these is the text 'The character set and its collation'.

```
CREATE SCHEMA `storage_accounting` DEFAULT CHARACTER SET utf8 ;
```

```
CREATE TABLE `storage_accounting`.`manufacturer` (  
  `manufacturer` VARCHAR(45) NOT NULL,  
  `site` VARCHAR(45) NULL,  
  PRIMARY KEY (`manufacturer`));
```

```
CREATE TABLE `storage_accounting`.`model` (  
  `model` VARCHAR(45) NOT NULL,  
  `manufacturer` VARCHAR(45) NOT NULL,  
  `capacity gb` INT NOT NULL,  
  `spindle_speed rpm` INT NULL,  
  `interface` VARCHAR(45) NULL,  
  PRIMARY KEY (`model`));
```

```
CREATE TABLE `storage_accounting`.`characteristic` (  
  `serial` INT NOT NULL,  
  `model` VARCHAR(45) NOT NULL,  
  `date` DATE NULL,  
  `failure` DATE NULL,  
  `comments` VARCHAR(45) NULL,  
  PRIMARY KEY (`serial`));
```

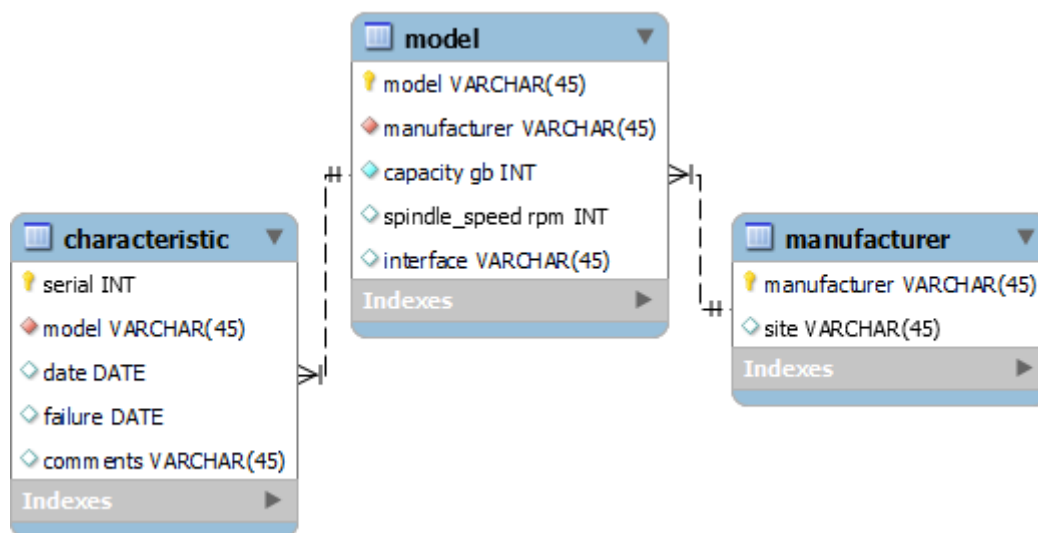
```

ALTER TABLE `storage_accounting`.`model`
ADD INDEX `mf_idx` (`manufacturer` ASC) VISIBLE;
;
ALTER TABLE `storage_accounting`.`model`
ADD CONSTRAINT `mf`
  FOREIGN KEY (`manufacturer`)
  REFERENCES `storage_accounting`.`manufacturer` (`manufacturer`)
  ON DELETE CASCADE
  ON UPDATE CASCADE;

ALTER TABLE `storage_accounting`.`characteristic`
ADD INDEX `md_idx` (`model` ASC) VISIBLE;
;
ALTER TABLE `storage_accounting`.`characteristic`
ADD CONSTRAINT `md`
  FOREIGN KEY (`model`)
  REFERENCES `storage_accounting`.`model` (`model`)
  ON DELETE CASCADE
  ON UPDATE CASCADE;

```

1. Графическая схема (выгруженная из MySQL Workbench)



2. Ссылка на борд в pastebin

<https://pastebin.com/18AU56aw>

3. Отдельно запросы по добавлению данных в таблицы.

```
INSERT INTO `storage_accounting`.`manufacturer` (`manufacturer`, `site`) VALUES ('m1', 'www.m1');
```

```
INSERT INTO `storage_accounting`.`manufacturer` (`manufacturer`, `site`) VALUES ('m2', 'www.m2');
```

```
INSERT INTO `storage_accounting`.`manufacturer` (`manufacturer`, `site`) VALUES ('m3', 'www.m3');
```

	manufacturer	site
▶	m1	www.m1
	m2	www.m2
	m3	www.m3

```
INSERT INTO `storage_accounting`.`model` (`model`, `manufacturer`, `capacity gb`, `spindle_speed rpm`, `interface`) VALUES ('m01', 'm1', 100, 1000, 'SATA');
```

```
INSERT INTO `storage_accounting`.`model` (`model`, `manufacturer`, `capacity gb`, `spindle_speed rpm`, `interface`) VALUES ('m02', 'm1', 256, 5400, 'SATA');
```

```
INSERT INTO `storage_accounting`.`model` (`model`, `manufacturer`, `capacity gb`, `spindle_speed rpm`, `interface`) VALUES ('m03', 'm2', 256, 7200, NULL);
```

```
INSERT INTO `storage_accounting`.`model` (`model`, `manufacturer`, `capacity gb`, `spindle_speed rpm`, `interface`) VALUES ('m04', 'm2', 512, 7200, NULL);
```

```
INSERT INTO `storage_accounting`.`model` (`model`, `manufacturer`, `capacity gb`, `spindle_speed rpm`, `interface`) VALUES ('m05', 'm3', 512, 5400, NULL);
```

	model	manufacturer	capacity gb	spindle_speed rpm	interface
▶	m01	m1	100	1000	SATA
	m02	m1	256	5400	SATA
	m03	m2	256	7200	NULL
	m04	m2	512	7200	NULL
	m05	m3	512	5400	NULL

```
INSERT INTO `storage_accounting`.`characteristic` (`serial`, `model`, `date`, `failure`, `comments`) VALUES (111, 'm01', '2020-02-02', '2022-02-02', NULL);
```

```
INSERT INTO `storage_accounting`.`characteristic` (`serial`, `model`, `date`, `failure`, `comments`) VALUES (222, 'm02', '2021-02-02', NULL, NULL);
```

```
INSERT INTO `storage_accounting`.`characteristic` (`serial`, `model`, `date`, `failure`, `comments`) VALUES (333, 'm03', '2022-01-01', NULL, NULL);
```

```
INSERT INTO `storage_accounting`.`characteristic` (`serial`, `model`, `date`, `failure`, `comments`) VALUES (444, 'm04', '2023-03-03', NULL, NULL);
```

```
INSERT INTO `storage_accounting`.`characteristic` (`serial`, `model`, `date`, `failure`, `comments`) VALUES (555, 'm05', '2024-01-01', NULL, NULL);
```

	serial	model	date	failure	comments
▶	111	m01	2020-02-02	2022-02-02	NULL
	222	m02	2021-02-02	NULL	NULL
	333	m03	2022-01-01	NULL	NULL
	444	m04	2023-03-03	NULL	NULL
	555	m05	2024-01-01	NULL	NULL

4. Продемонстрировать функцию по сохранению целостности БД (на примере добавления, изменения, удаления данных) т.е. показать, что при добавлении, возможно добавить новую строку в связанную сущность и т.д.

Для этого удалим запись из таблицы manufacturer со значением m1, что должно привести к удалению связанных записей из таблиц model и characteristics.

```
DELETE FROM storage_accounting.manufacturer WHERE manufacturer = 'm1';
```

	manufacturer	site
▶	m2	www.m2
	m3	www.m3

	serial	model	date	failure	comments
▶	333	m03	2022-01-01	NULL	NULL
	444	m04	2023-03-03	NULL	NULL
	555	m05	2024-01-01	NULL	NULL

	model	manufacturer	capacity gb	spindle_speed rpm	interface
▶	m03	m2	256	7200	NULL
	m04	m2	512	7200	NULL
	m05	m3	512	5400	NULL

Обновим запись из таблицы model со значением m03, что должно привести к обновлению связанных записей из таблицы characteristics.

```
UPDATE `storage_accounting`.`model` SET `model` = 'mmm' WHERE (`model` = 'm03');
```

	model	manufacturer	capacity gb	spindle_speed rpm	interface
▶	m04	m2	512	7200	NULL
	m05	m3	512	5400	NULL
	mmm	m2	256	7200	NULL

	serial	model	date	failure	comments
▶	333	mmm	2022-01-01	NULL	NULL
	444	m04	2023-03-03	NULL	NULL
	555	m05	2024-01-01	NULL	NULL

Попытаемся добавить в таблицу characteristics диск с несуществующей моделью.

```
INSERT INTO `storage_accounting`.`characteristic` (`serial`, `model`, `date`, `failure`, `comments`) VALUES (000, 'Unknown', '2024-04-04', NULL, NULL)
```

	serial	model	date	failure	comments
▶	333	mmm	2022-01-01	NULL	NULL
	444	m04	2023-03-03	NULL	NULL
	555	m05	2024-01-01	NULL	NULL

Ограничения целостности не позволяют это сделать.

Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails
 (`storage\_accounting`.`characteristic`, CONSTRAINT `md` FOREIGN KEY (`model`) REFERENCES `model` (`model`) ON DELETE CASCADE ON UPDATE CASCADE)